Airline Data Analytics Project

SQL • MySQL Workbench • Excel Dashboard

Project overview

This project analyzes airline bookings, passengers, flight operations, routes, ticket classes, and revenue using a relational SQL database. The supplied records are synthetic sample data created for learning and portfolio demonstration.

Business objectives

• Measure passenger, flight, and booking volumes.

• Compare airline revenue and operational activity.

• Identify popular routes and ticket classes.

• Track flight status, including cancelled flights.

• Review monthly booking and revenue patterns.

Dataset inventory

Table	Rows	Purpose
Airlines	5	Airline names and home countries
Passengers	150	Passenger demographics
Flights	120	Routes, dates, airline, and status
Bookings	500	Booking date, ticket class, and fare

Key sample KPIs

KPI	Sample result
Passengers	150
Flights	120
Bookings	500
Booking revenue	■4,169,273.63
Cancelled flights	10
Cancellation rate	8.3%

SQL skills demonstrated

CREATE DATABASE, CREATE TABLE, primary and foreign keys, SELECT, WHERE, JOIN, GROUP BY, HAVING, ORDER BY, aggregate functions, date functions, CASE expressions, common table expressions, and window functions (DENSE_RANK and running totals).

Dashboard contents

The Excel workbook includes raw data sheets, a summary sheet with grouped analysis, KPI cards, and charts for revenue by ticket class, monthly bookings, revenue by airline, and flight status distribution. Summary tables are PivotTable-style aggregations; refreshable native Excel PivotTables can be added in Excel if required.

How to use

1. Open Airline_Analytics_SQL.sql in MySQL Workbench and run the database/table creation section.

2. Import Airlines.csv, Passengers.csv, Flights.csv, and Bookings.csv into the matching tables using the Table Data Import Wizard.

3. Run the analysis queries in the SQL file.

4. Open Airline_Analytics_Dashboard.xlsx in Excel and review the Dashboard and Summary sheets.

5. Use the PDF report as project documentation and customize the findings before publishing to your portfolio.

Important note

All names, flights, bookings, dates, and amounts in this project are synthetic and are intended for practice only. Results are illustrative and should not be presented as real airline performance.